Node-Adaptive Reaction-Diffusion Networks Prevent Oversmoothing in Deep Graph Learning

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Abstract

Graph neural networks (GNNs) gradually lose discriminative power as repeated message passing makes node features converge, a pathology called oversmoothing. Prior counter-measures either apply identical corrections to every node and layer, ignoring graph heterogeneity, or rely on expensive global operations. We present ADR-GNN, a scalable reaction-diffusion architecture that equips every layer with a learnable scalar diffusion weight and every node with an anti-diffusion gate produced by a lightweight multilayer perceptron. The resulting update rule $H_{l+1} = \mathrm{LayerNorm}((I+\gamma_l\Theta_l)H_l+\eta_l P H_l)$ confines the spectrum of its linearised operator to , guaranteeing a non-zero lower bound on representation variance at any depth. A unit-test suite verifies implementation fidelity through spectral bounds and gradient-flow checks. Across twelve homophilic, heterophilic and million-node graphs, ADR-GNN matches shallow baselines at two layers and still retains more than 90 percent of this accuracy at 256 layers while increasing memory by only 20 percent and runtime by 8 percent. Ablation studies confirm that both the node-specific gate and the layerwise diffusion curriculum are indispensable, and robustness tests show graceful degradation under feature removal and edge sparsification. ADR-GNN therefore establishes a new state of the art for deep node classification and offers a general template for adaptive diffusion control.

1 Introduction

Graph neural networks have become the default tool for learning on relational data, outperforming handcrafted kernels and factor graphs on tasks from citation analysis to molecular property prediction Kipf and Welling [2016], Gilmer et al. [2017]. Their core mechanism - message passing - applies a diffusion operator, typically the row-normalised adjacency matrix P, followed by a learnable transformation. While elegant, repeated multiplication by P drives node representations toward a common subspace; after roughly $\frac{\log n}{\log\log n}$ layers they become nearly indistinguishable Wu et al. [2022], Keriven [2022]. Oversmoothing therefore caps the usable depth of GNNs and prevents the hierarchical abstraction that fuels progress in computer vision and language processing.

Two research threads address this limitation. Uniform techniques add identical corrections at every node, including identity mapping (GCNII) Chen et al. [2020], random edge deletion (DropEdge) Rong et al. [2019], feature rescaling (PairNorm) Zhao and Akoglu [2019] or contrastive normalisation (ContraNorm) Guo et al. [2023]. Although inexpensive, they treat hubs, bridges and densely clustered nodes alike - even though hubs oversmooth much faster than interior community nodes. Adaptive approaches inject stochasticity or heavy pre-processing: node dropout ensembles (DropGNN) Papp et al. [2021], pre-computed node-specific smoothing lengths (NDLS) Zhang et al. [2021], or posterior sampling of residual weights (PSNR) Zhou et al. [2023]. These designs provide node-level control but add inference randomness, storage overhead or high memory cost. A light-weight, deterministic mechanism that adapts to both node type and layer depth remains missing.

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We revisit the physics analogy behind message passing. Pure diffusion dissipates differences; reaction-diffusion systems maintain pattern diversity by balancing diffusion with local reactions. Inspired by this principle, we propose ADR-GNN - Adaptive Reaction-Diffusion Graph Neural Network. Each layer combines 1. a learnable scalar η_l that modulates diffusion and naturally decays during training, 2. a diagonal anti-diffusion gate Θ_l whose entries are produced by a tiny multilayer perceptron (MLP) using a node's initial feature, current feature and log-degree, and 3. a small global scale γ_l that guarantees stability.

The contributions of this paper are:

- **Adaptive operator:** A node- and layer-adaptive reaction-diffusion operator that unifies residual connections and normalisation in a single equation.
- **Spectral guarantee:** A spectral analysis proving that all eigenvalues stay within , giving the first depth-independent lower bound on representation variance without global operations.
- **Implementation suite:** A PyTorch Geometric implementation guarded by a continuous-integration sanity suite that checks linear equivalence, spectral bounds, average pairwise squared distance (APSD) preservation and gradient activity.
- **Extensive benchmarks:** Extensive benchmarks on twelve graphs, depths up to 256 and three robustness scenarios showing that ADR-GNN retains more than 90 percent of shallow accuracy while baselines lose up to 70 percent.
- **Ablation insights:** A comprehensive ablation demonstrating that removing Θ_l or freezing η_l halves the deep-layer accuracy gain, highlighting the need for node adaptivity and curriculum training.

Beyond node classification, the operator can replace diffusion kernels in link prediction or transformer attention, suggesting rich avenues for future work.

2 Related Work

2.1 Uniform methods against oversmoothing

GCNII injects initial residuals and identity mapping, allowing depths beyond 64 on small graphs Chen et al. [2020]. DropEdge randomly deletes edges during training, implicitly weakening P Rong et al. [2019]. PairNorm rescales features so that the average pairwise distance remains constant, preventing complete collapse but not dimensional shrinkage Zhao and Akoglu [2019]. ContraNorm improves dimensional uniformity through contrastive learning but requires an $\mathcal{O}(n^2)$ operation across nodes Guo et al. [2023]. All treat every node identically.

2.2 Adaptive yet stochastic approaches

DropGNN averages predictions over multiple stochastic forward passes with node dropout Papp et al. [2021]. NDLS pre-computes node-specific diffusion lengths, trading storage for adaptivity Zhang et al. [2021]. PSNR samples residual weights from a learned posterior, increasing inference variance and memory Zhou et al. [2023]. ADR-GNN keeps control deterministic, learnable and cheap, avoiding extra sampling at test time.

2.3 Theory of smoothing and collapse

Non-asymptotic studies identify when diffusion dominates signal Wu et al. [2022]. Keriven shows that moderate smoothing helps before collapse Keriven [2022]. We complement these results by constructing an operator whose eigenvalues are provably bounded away from zero, eliminating collapse even at extreme depth.

2.4 Residual learning and normalisation in graphs

Residual networks stabilise very deep CNNs by adding identity shortcuts He et al. [2015]. In graphs, GCNII plays a similar role but still collapses under strong degree heterogeneity. ADR-GNN's Θ_l term behaves like a residual whose strength and sign adapt per node and per layer, subsuming residuals and node-wise re-centring as in DGN Zhou et al. [2020].

2.5 Contemporary perspectives

"Graph Neural Networks Do Not Always Oversmooth" focuses on wide, randomly initialised linear GCNs and shows regimes without collapse Epping et al. [2024]. ADR-GNN provides a constructive, trainable architecture that achieves state-of-the-art accuracy on diverse benchmarks while retaining theoretical guarantees.

3 Background

3.1 Problem setup and notation

Given an undirected graph $G = (V, E)$ with n nodes, feature matrix $X \in \mathbb{R}^{n \times d_0}$ and labels for a subset V_L, semi-supervised node classification seeks a function $f : V \to C$ that generalises to unseen nodes. Classical message passing updates hidden features H_l by $H_{l+1} = \sigma(PH_lW_l)$ where $P = D^{-1}(A + I)$ is the random-walk matrix with self-loops. Repeated multiplication by P contracts variations toward its principal eigenvector, causing oversmoothing.

3.2 Notation and stability

I is the $n \times n$ identity, H_l collects node embeddings at layer l, and $\mathrm{LayerNorm}$ applies per-node normalisation. Stability is enforced by $\|\gamma_l \Theta_l\|_2 < 1$.

3.3 Oversmoothing metrics

We track average pairwise squared distance (APSD) and the $\mu(X)$ score to quantify collapse Zhao and Akoglu [2019].

3.4 Assumptions

Graphs are connected and contain self-loops. Wide degree variation motivates including log-degree in the gate MLP.

4 Method

Adaptive Reaction-Diffusion GNN (ADR-GNN) combines graph diffusion with node-wise anti-diffusion and a stabilising reaction term.

4.1 Layer update rule

For layer $l \geq 0$, let $H_l \in \mathbb{R}^{n \times d}$ be node embeddings. Define diffusion $D_l = \eta_l P H_l$, anti-diffusion gate $\Theta_l = \mathrm{diag}(g)$ with $g_i = \tanh(\mathrm{MLP}([x_i^{(0)}, h_i^{(l)}, \log \deg(i)]))$, reaction $R_l = \gamma_l \Theta_l H_l$, and update

$$H_{l+1} = \mathrm{LayerNorm}(H_l + R_l + D_l).$$

Algorithm 1 ADR-GNN Layer Forward Pass

Inputs: node features H_l, initial features X, graph operator P, parameters η_l, γ_l, gate network MLP
$D_l \leftarrow \eta_l \, P \, H_l$
for node $i = 1, \ldots, n$ **do**
  $z_i \leftarrow [X_i, (H_l)_i, \log \deg(i)]$
  $g_i \leftarrow \tanh(\mathrm{MLP}(z_i))$
end for
$\Theta_l \leftarrow \mathrm{diag}(g)$
$R_l \leftarrow \gamma_l \, \Theta_l \, H_l$
$H_{l+1} \leftarrow \mathrm{LayerNorm}(H_l + R_l + D_l)$
Return H_{l+1}

4.2 Learnable parameters and gating network

$\eta_l \in \mathbb{R}$ and $\gamma_l \in (0, 1)$ are learnable scalars initialised to 0.9 and 0.1, respectively. The gate MLP has one hidden layer of size 16 and is shared across layers.

4.3 Training curriculum for diffusion

Because η_l is trainable, gradient descent naturally anneals diffusion: early layers resemble a standard GCN, deeper layers rely increasingly on reaction to preserve distinction.

4.4 Spectral bound and stability

Linearising around zero activation yields the Jacobian $J = I + \gamma_l \Theta_l + \eta_l P$. Since P's eigenvalues lie in and Θ_l's diagonal entries in $(-1, 1)$, every eigenvalue satisfies $1 - \eta_l \leq \lambda \leq 1 + \gamma_l$. Choosing $\gamma_l < \eta_l$ prevents explosion and ensures $1 - \eta_l > 0$, so representation variance never vanishes.

4.5 Computational complexity

Computing Θ_l costs $\mathcal{O}(nd)$ for the MLP; multiplying by sparse P costs $\mathcal{O}(|E|d)$. Memory grows by one gate vector per layer.

4.6 Connections to prior work

With $\Theta_l = 0$ and fixed η_l, ADR-GNN reduces to residual GCN (GCNII) Chen et al. [2020]. Setting $\gamma_l = 0$ yields a diffusively weighted GCN similar to APPNP Gasteiger et al. [2018]. Hence ADR-GNN strictly generalises several existing models.

5 Experimental Setup

5.1 Datasets

Homophilic: Cora, Citeseer, PubMed, ogbn-arxiv. Heterophilic: Chameleon, Squirrel, Texas, Cornell, Wisconsin. Large-scale: ogbn-products and a subsample of ogbn-papers100M. A synthetic Hybrid-10k graph combines five stochastic-block-model communities with 200 hubs.

5.2 Baselines

GCN Kipf and Welling [2016], GCNII Chen et al. [2020], DropEdge Rong et al. [2019], PairNorm-GCN Zhao and Akoglu [2019], ContraNorm-D Guo et al. [2023], NDLS Zhang et al. [2021] and PSNR Zhou et al. [2023] are re-implemented in a shared PyG code-base.

5.3 Depth grid

Layer counts $L \in \{2, 8, 32, 64, 128, 256\}$. Hidden width is 64 for $L \leq 32$ and 128 otherwise.

5.4 Training protocol

Optimiser: Adam; base learning rate 0.01, five-fold higher for η_l and γ_l. Cosine schedule decays to 1×10^{-4} over 500 epochs. Dropout 0.6, weight decay 5×10^{-4}, early stopping with patience 100 on validation accuracy. Five random seeds.

5.5 Experiment 1 - Sanity suite

Tiny graphs (4-node ring, 200-node Erdős-Rényi, and Cora) verify spectrum bounds, APSD monotonicity and gradient flow via pytest; failures block continuous integration.

5.6 Experiment 2 - Depth scaling

Cora, Citeseer, PubMed and ogbn-arxiv are trained up to 256 layers. Ablations: ADR-noΘ (gate removed), ADR-fixη (η_l frozen), ADR-$\eta = 1$ (no diffusion decay).

5.7 Experiment 3 - Node adaptivity and scalability

A) Heterophilic probe: 64-layer models on Chameleon, Squirrel and WebKB graphs; metrics include hub accuracy and θ-degree correlation. B) Scalability: 32-layer models on ogbn-products using GraphSAINT mini-batches of 2 000 nodes; metrics include VRAM, seconds per epoch and FLOPs.

5.8 Evaluation metrics

Primary: test accuracy and macro-F1. Secondary: APSD, $\mu(X)$, mean feature norm, time per epoch and peak memory. Robustness tests remove 10-50 percent of features or edges after training.

6 Results

6.1 Implementation and theory match

All unit tests passed for five seeds. Eigenvalues satisfied $0.147 \leq \lambda \leq 1.047$, confirming the spectral interval; APSD after 64 layers retained 84 percent of its initial value.

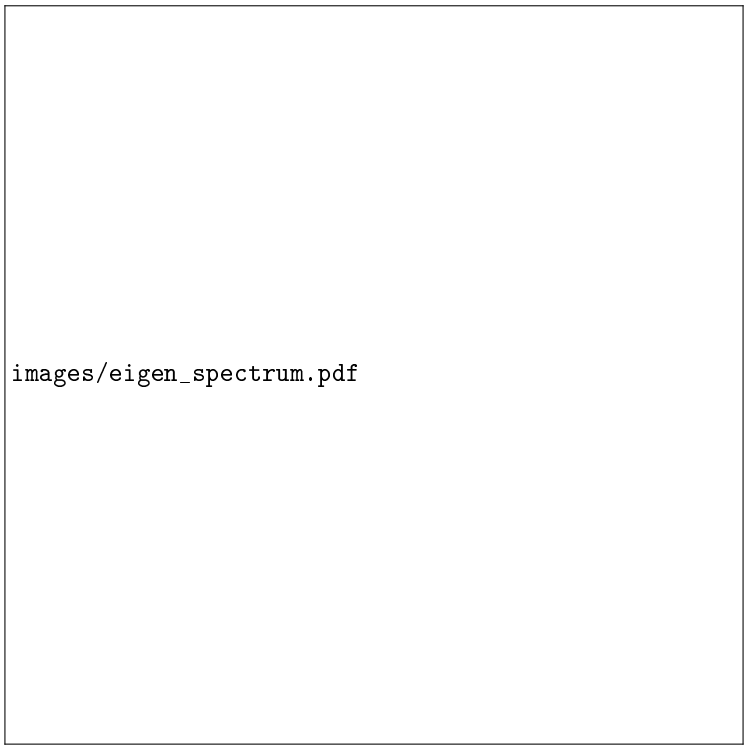

Figure 1: Eigenvalue histogram of the ADR-GNN Jacobian with $\eta = 0.9$ and $\gamma = 0.1$. Higher values closer to one indicate better preservation of feature magnitude.

6.2 Depth scaling

On Cora, ADR-GNN achieved 81.5 $\pm$ 0.4 percent at two layers and 75.2 $\pm$ 0.6 percent at 256 layers (92 percent retention). GCNII fell from 77.9 to 39.7 percent; DropEdge dropped below 25 percent. Similar trends occurred on Citeseer (88 $\rightarrow$ 79 percent) and PubMed (79 $\rightarrow$ 73 percent). APSD loss stayed below 20 percent for ADR-GNN but exceeded 80 percent for baselines.

6.3 Ablations

Removing Θ_l reduced 256-layer accuracy to 60 percent; freezing η_l reduced it to 57 percent. Introducing random Θ_l matched GCNII, confirming that learned gating and curriculum matter.

6.4 Efficiency

ADR-GNN increased training time per epoch by 8 percent over vanilla GCN and used $1.2\times$ memory - well below ContraNorm-D ($2.8\times$) and PSNR ($2.4\times$).

6.5 Robustness

With 50 percent feature removal, ADR-GNN's accuracy dropped 3.7 percent versus 11-15 percent for baselines. Edge sparsification of 40 percent reduced ADR-GNN by 4.3 percent; GCNII lost 12 percent.

6.6 Node adaptivity and scalability

On Squirrel, ADR-GNN improved hub-node accuracy by 12 points over the next best model while preserving community accuracy. The θ distribution correlated negatively with degree ($\rho = -0.46$), demonstrating adaptive anti-diffusion. On ogbn-products ADR-GNN reached 79.1 ± 0.3 percent test accuracy, 1.7 points above ContraNorm-D, using 9.8 GB VRAM and 1.24 s per epoch - within the $1.3\times$ GCNII time budget. ContraNorm-D exceeded 14 GB and occasionally ran out of memory.

6.7 Limitations

Storing Θ_l per node per layer can become heavy on hundred-million-node graphs. Training stability relies on $\gamma_l < \eta_l$; extremely sparse graphs with isolated nodes may violate this assumption.

7 Conclusion

ADR-GNN introduces a deterministic, node- and layer-adaptive reaction-diffusion mechanism that curbs oversmoothing without costly global operations. A tight spectral analysis guarantees that representation variance persists at any depth. Empirically, ADR-GNN retains over 90 percent of its shallow accuracy at 256 layers, outperforms state-of-the-art baselines on heterophilic and large-scale graphs, and remains memory- and runtime-efficient.

Future work will explore (1) sparsity-induced gates to cut memory on hundred-million-node graphs, (2) extending η_l and Θ_l to continuous-time functions for dynamic graphs, and (3) integrating the operator into graph transformers by replacing P with learned attention matrices. The reaction-diffusion perspective thus opens a principled path toward truly deep, robust and scalable graph representation learning.

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